nature communications

Article

https://doi.org/10.1038/s41467-026-72926-3

Aerosol effective radiative forcing increases Earth's energy imbalance in recent decades

Received: 15 October 2025

Accepted: 23 April 2026

Published online: 11 May 2026

Check for updates

Tianle Yuan[1,2] ✉, Hua Song[1,3], Ryan Kramer[4], Lazaros Oreopoulos[1], Shiv Priyam Raghuraman[5], Robert Wood[6] & Mian Chin[2]

Earth's energy imbalance (EEI), a key driver of climate change, has risen markedly over the last two decades. Greenhouse gas forcing, aerosol forcing, and cloud feedback all contribute to this increase. However, the role of aerosol forcing, particularly effective radiative forcing through aerosol-cloud interactions (ERF_{ACI}), remains highly uncertain and closely intertwined with cloud feedback. Here, we estimate shortwave ERF_{ACI} using satellite observations and show its importance as a contributor to the EEI increase between 2003 and 2023. ERF_{ACI} exhibits a significant increasing trend of 0.33 Wm^{-2}/*decade* averaged over oceans between 60°S and 60°N. The increasing trend of ERF_{ACI} stems from a global decline in cloud droplet number concentration driven by decreasing anthropogenic aerosol emissions. It is of similar magnitude as the combined instantaneous forcing trend from greenhouse gases and aerosol-radiation interactions. Our results may reduce the gap between the simulated and observed shortwave contribution to EEI trends while implying a weak shortwave cloud feedback. They also suggest a stronger aerosol forcing than the multi-model mean.

The Earth's energy imbalance (EEI) is the difference between absorbed solar radiation and emitted outgoing longwave radiation[1,2]. A positive EEI indicates an Earth that is gaining heat and leads to higher global mean temperature, sea level rise, and melting of ice and snow. EEI is the net effect of all processes that affect the Earth's energy budget. Reduced outgoing longwave radiation by greenhouse gases increases EEI and heats up the atmosphere and ocean, which triggers changes in variables such as global mean surface temperature, cloud radiative effect, water vapor, and surface albedo which in turn affect EEI (see "Methods")[3,4]. Anthropogenic aerosols can affect EEI through direct changes to the absorbed solar radiation, i.e., the net solar flux into the climate system, and indirectly modifying cloud properties, a combined effect known as total effective radiative forcing (ERF) from aerosols. The magnitude of total aerosol forcing is potentially large, but remains highly uncertain[5,6]. Internal climate variations, and changes in natural aerosol and solar irradiance can also affect EEI, but their long-term contribution is much smaller[1–3,7].

An estimated 90% of EEI goes toward increased ocean heat content, while the remainder melts snow and ice and increases land and atmosphere temperature[2,3,8]. Measurements of ocean heat content show a consistent positive trend since the 1960s, and an accelerated trend during the last two decades[9]. Satellite observations from NASA's Clouds and the Earth's Radiant Energy System (CERES) instrument[10] confirm this positive trend of 0.63 Wm^{-2}/decade for the period 2003–2023 over global oceans between 60°S and 60°N (0.51 Wm^{-2}/decade for the whole globe)[3,11]. Together, these independent records provide a top-down constraint on the combined effect of forcing, feedback, and internal variability in shaping Earth's evolving energy budget.

Understanding the drivers of EEI and its temporal evolution is critical for projecting shifts in decadal change, hydrological cycle and

[1]Earth Sciences Division, NASA Goddard Space Flight Center, Greenbelt, MD, USA. [2]GESTAR-II, University of Maryland, Baltimore, MD, USA. [3]Science & Technology Corporation, Columbia, MD, USA. [4]Geophysical Fluid Dyanmics Laboratory, NOAA Department of Commerce, Princeton, NJ, USA. [5]Department of Climate, Meteorology & Atmospheric Sciences, University of Illinois, Urbana, IL, USA. [6]Department of Atmospheric Sciences, University of Washington, Seattle, WA, USA. ✉e-mail: tianle.yuan@nasa.gov

https://doi.org/10.1038/s41467-026-72926-3

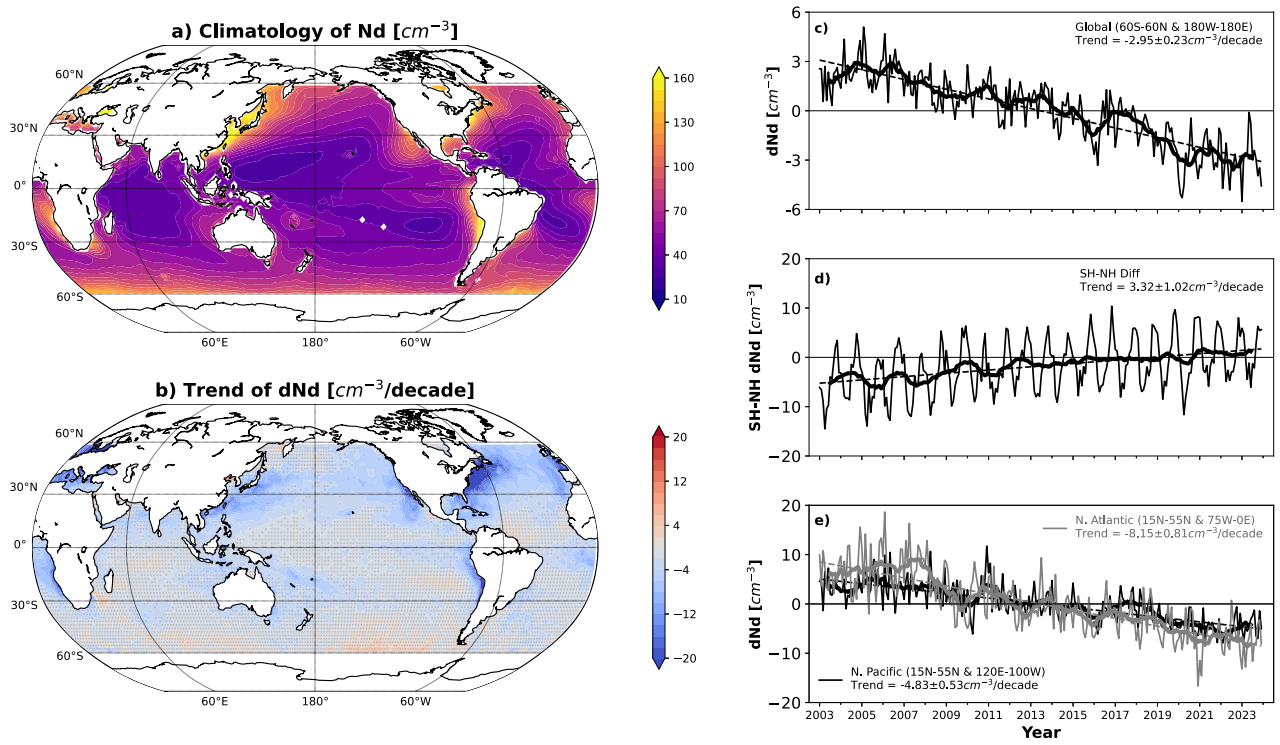


Fig. 1 | Observed MODIS N_d climatology and trends for different regions. a Aqua MODIS derived N_d climatology over the ocean for 2003–2023. **b** Linear trend in N_d for the same period; **c** time series of global ocean mean N_d and its 12-month running mean (in black); **d** time series of hemispheric difference in N_d, southern hemisphere (SH) minus northern hemisphere (NH); **e** time series of regional mean N_d over the North Pacific and the North Atlantic (**e**) basins. Stippled regions in panel b are for $p > 0.05$[57].

the planet's energy flows[10]. By integrating high-quality reanalysis data, satellite observations, and modeling tools, EEI and its longwave and shortwave components, can be decomposed to quantify different contributing processes (see Methods). For example, partial radiative perturbation analysis demonstrates that clouds, water vapor, trace gases, surface albedo changes, and global mean temperature all affect EEI trends[11,12]. Natural variability alone cannot explain the recent increase in EEI[7]. After removing the effect of feedbacks and ERF_{ACI}, the combined instantaneous forcing from greenhouse gases and aerosol direct effect contributes about 0.33 Wm^{-2}/decade to EEI between 2003 and 2023 using updated data from ref. 13.

The radiative impact of clouds plays a major role in increasing EEI[3,11]. However, separating the effects of cloud feedback—temperature-mediated radiative response -[6,7,14] from ERF_{ACI} remains difficult because both are uncertain and intertwined. Cloud feedback remains a key uncertainty in climate feedback studies, while ERF_{ACI} is the leading source of uncertainty in radiative forcing estimates[5,6]. Both processes contribute to changes in the cloud radiative effect (see Methods). Consequently, improving the estimate of either can help refine the other, given the total cloud radiative effect as a constraint[11,13,15]. For example, multi-model mean estimates of ERF_{ACI} suggest a small total cloud feedback between 2001 and 2020 under the constraint of adjusted CERES cloud radiative effect (aCRE), i.e., the CRE after accounting for cloud masking effects:

$$\mathrm{aCRE} = CRE - CM = ERF_{\mathrm{ACI}} + CRR + ERF_{GHGSW} + \epsilon, \quad (1)$$

where CRE is cloud radiative effect from CERES observations, CM is cloud masking effect that can be calculated[13], CRR is the temperature mediated cloud radiative response, i.e., the true cloud feedback term. ERF_{GHGSW} are ERF due to cloud adjustments to greenhouse gas forcing in the shortwave.

An alternative approach is to derive ERF_{ACI} trend over the past two decades based on observations. This is now possible through well-calibrated, long-term satellite observations of clouds and aerosols and reanalysis data, along with recent methodological advances in obtaining observed ERF_{ACI}[16–20]. NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) instrument provides global retrievals of cloud variables such as cloud droplet effective radius, cloud optical depth, and cloud fraction[21]. At the same time, opportunistic experiments of aerosol-cloud interactions offer quantitative characterization of cloud adjustments to aerosols as a function of background conditions[22,23]. Such cloud adjustment functions can be used to build bottom-up estimates of ERF_{ACI} that are calibrated against independent estimates from top-down approaches and modeling studies (see Methods)[19,24,25]. Because the calibration of Aqua MODIS is remarkably stable[26], trends derived from it capture actual changes rather than instrumental drifts, before the satellite orbit itself started drifting. Moreover, anthropogenic aerosol emissions and aerosol loading have undergone substantial regional and global changes during the same period (see Figs. S1 and S2)[27]. Changes in aerosol emissions affect cloud formation and leave detectable trends in cloud droplet number concentration (N_d) at both regional and global scales (Figs.1 and S2)[28], providing an opportunity to study the resulting ERF_{ACI} trend.

This study quantifies the ERF_{ACI} trend between 2003 and 2023 using observations, assesses its importance relative to other processes in the context of EEI increase, and discusses its implications. Most results reported here pertain to global oceans between 60°S and 60°N because our cloud adjustment functions have been derived from samples within this domain (see Methods for more details)[23]. Equivalent global means are obtained by assuming no ERF_{ACI} trend or only the Twomey effect in the remainder of the globe (see "Methods").

Results

MODIS-based N_d trend

Figure 1a shows the MODIS climatology map of N_d in 60°S to 60°N oceans for the last two decades (2003–2023). The climatological mean over this domain is 62 cm^{-3}. Notably larger N_d is found downwind of major economic centers such as East Asia, North America, and Europe[28,29]. Oceanic areas to the west of major continents also have elevated N_d, compared to remote oceans, due to both anthropogenic and natural emissions such as wild fires. For example, N_d is elevated off ths coasts of California and Baja California due to a combination of emissions from anthropogenic sources, fires, and shipping while elevated values off ths coasts of Peru and Chile are due to ocean upwelling and anthropogenic and shipping emissions[30]. The corresponding N_d trend map is shown in Fig. 1b. Far away from coastal areas trends are flat, or positive (but often not statistically significant) in several regions of the Indian Ocean, Southern Ocean and Tropical Atlantic. The cause of these increasing trends in remote oceans is not yet known, but may be due to increases in natural sources[31] considering that areas downstream of major economic centers and off west coast of major continents where anthropogenic effects are far larger, exhibit varying degrees of N_d negative trends.

Overall, we observe a decreasing trend of −2.95 ± 0.23 cm^{-3}/decade in the global ocean mean of N_d (Fig. 1c), or roughly 5%/decade in relative terms[28]. The strongest downward trend is observed off the east coast of the US, peaking at above −20 cm^{-3}/decade. In other areas the maximum negative trend is close to −10 cm^{-3}/decade (Fig. 1b). The mean decrease rates for the north Pacific and Atlantic basins are −4.83 ± 0.53 cm^{-3}/decade and **-8.15** ± 0.81 cm^{-3}/decade, respectively (Fig. 1e). Such downward trends are consistent with regional trends of aerosol emissions (Figs. S2 and ref. 28) as well as decreasing aerosol optical depth from MODIS (Fig. S2). An exception is the Indian Ocean where N_d shows a weak and statistically insignificant positive trend over some areas (Fig. 1b), which while consistent with increasing emissions from India, is weaker than expected given the substantial increase in aerosol emissions and observed aerosol optical depth. A strong contrast between N_d trends in the northern (NH) and southern (SH) hemispheres can also be seen (Fig. 1d). The linear trend in the N_d difference between SH and NH is 3.32 ± 1.02 cm^{-3}/decade, most of which being driven by the anthropogenic decrease in the NH. The SH-NH difference in N_d has disappeared or even become positive since 2015 (thick line in Fig. 1d).

ERF_{ACI} trend and its impact

Figure 2 shows the 2003–2023 time series of anomalies in CERES EEI (2a), ERF_{ACI} (2b), the combined instantaneous forcing of aerosol-radiation interactions (ARI) and greenhouse gases (2c), adjusted cloud radiative effect (aCRE) (2d), i.e., CERES shortwave cloud radiative effect minus cloud masking effects (see Methods), and global mean temperature over oceans between 60°S and 60°N (2e). CERES EEI and SW aCRE are increasing at **0.63** ± 0.25 Wm^{-2}/decade and **0.77** ± 0.25 Wm^{-2}/decade, respectively. ERF_{ACI} estimates are based on observed N_d anomalies from Aqua MODIS combined with observations of cloud albedo, cloud adjustment functions, and reanalysis data (see Methods for details)[16,23]. This means that the calculated ERF_{ACI} includes effects from both anthropogenic and natural aerosol changes, but we assume decadal changes in N_d are mainly driven by anthropogenic changes. The calculated positive trend in ERF_{ACI} mostly reflects the impact of decreasing anthropogenic emissions. The N_d trend map (Fig. 1b), indicates that the background N_d over remote oceans is actually increasing slightly, which likely reflects an overall increase in natural sources (Figs. 1, S2, and S3) and makes our estimate of ERF_{ACI} conservative since the observed increase in natural aerosols has a cooling effect.

The calculated total shortwave ERF_{ACI} exhibits a linear trend of **0.33** ± 0.03 Wm^{-2}/decade during this period averaged over global oceans between 60°S and 60°N, representing 53% of the observed postive EEI trend over the same region. In comparison, the combination of instantaneous radiative forcing from greenhouse gases and aerosol direct effects amounts to **0.33** ± 0.05 Wm^{-2}/decade (see Fig. 2c and ref. 13). The contribution of ERF_{ACI} is thus similar to the combined forcing from greenhouse gases and aerosol-radiation interactions (ARI) to the EEI trend. It is important to note that ERF from greenhouse gases includes stratospheric and tropospheric adjustments (e.g., clouds) and is thus larger than their instantaneous forcing[6]. Taking into account these adjustments, the total effective forcing of greenhouse gases and aerosol-radiation interactions has a stronger trend than ERF_{ACI}[7]. It is also important to note that while ERF_{ACI}, forcing from greenhouse gases, and ARI drive the EEI increase, different feedback processes and, to a lesser extent, natural variability can also reduce or amplify the radiative forcing[11,13,15].

Our calculated ERF_{ACI} trend represents 43% of the observed trend in shortwave aCRE. The remainder is the combined effect of cloud feedback and cloud adjustments to greenhouse gas forcing (see equation 1)[13,15,32]. Our estimate finds ERF_{ACI} contributing more to the aCRE trend than cloud adjustments to greenhouse gas forcing and cloud feedback compared to model-based estimates[15,32]. For example, the modeled ERF_{ACI} and cloud adjustment to greenhouse gas forcing contribute less than half to the trend in shortwave aCRE[15] while our calculation of ERF_{ACI} alone accounts close to half of aCRE trend.

If we scale our ERF_{ACI} value by the fraction of the planet's area occupied by oceans within (60°S ~ 60°N) i.e., assuming no ERF_{ACI} trend elsewhere, its global mean value is **0.21** ± 0.02 Wm^{-2}/decade (See Table 1). Global model simulations capture the increasing trend of EEI in recent decades, but with a discrepancy of 0.1 Wm^{-2}/decade between multi-model mean and CERES observed EEI trends[7,27,33,34]. The EEI increase is dominated by the SW heating trend that is partially compensated by the LW cooling trend although models underestimate the LW trend[3,15,27]. Observation-based radiative kernel calculations can explain the LW component of EEI quite well[13], leaving the SW component as the main component to be explained observationally. The multi-model mean estimate of ERF_{ACI} trend is + 0.06 Wm^{-2}/decade in the SW[15]. A subset of models show a total SW aerosol ERF trend of + 0.25 Wm^{-2}/decade for 2001–2019[27]. Three out of four models in this subset have among the strongest historical ERF_{ACI}[35], suggeting these are highly sensitive models to aerosol perturbations. Removing the trend of IRF from ARI at + 0.06 Wm^{-2}/decade[13], we estimate the modeled ERF_{ACI} trend to be around + 0.15 Wm^{-2}/decade in ref. 27. Our scaled ERF_{ACI} trend adds between +0.06 to +0.15 Wm^{-2}/decade to the multi-model EEI trend in the SW. If we add the forcing due to Twomey effect over land and assume total LWP and CF cloud adjustment effect to be zero, the total global mean ERF_{ACI} would be + 0.25 Wm^{-2}/decade. This is 0.1 Wm^{-2}/decade higher than the more sensitive models[27] and would reduce the trend difference of + 0.2 Wm^{-2}/decade in the SW by 50% (see Table 1). Our analysis suggests that models with higher total aerosol ERF or ERF_{ACI} are more consistent with our observational estimates[15,27].

We follow[3] and examine the temporal evolution of these variables during three sub-periods, namely the hiatus period (Jan 2003-May 2010), transition to El Nino (Jun 2010–May 2016), and post El Nino (Jun 2016–Dec 2023). During the hiatus period, the surface temperature trend is flat (Fig. 2e) while the combined instantaneous forcing from greenhouse gas and aerosol total effects has a statistically insignificant trend of −0.05 ± 0.20 Wm^{-2} (Fig. 2c). During the transition to El Nino sub-period, both ERF_{ACI} (Fig. 2b) and the combined forcing from greenhouse gases and ARI (Fig. 2c) produce a strong increasing trend, with the latter experiencing a much larger change. The trend of SW aCRE is especially strong at 1.83 ± 0.96 Wm^{-2}/decade (Fig. 2d), nearly doubling the total instantaneous forcing from greenhouse gases and total aerosol effects. This trend stands out among these three sub-periods and implies a strong contribution from cloud feedback and/or

https://doi.org/10.1038/s41467-026-72926-3

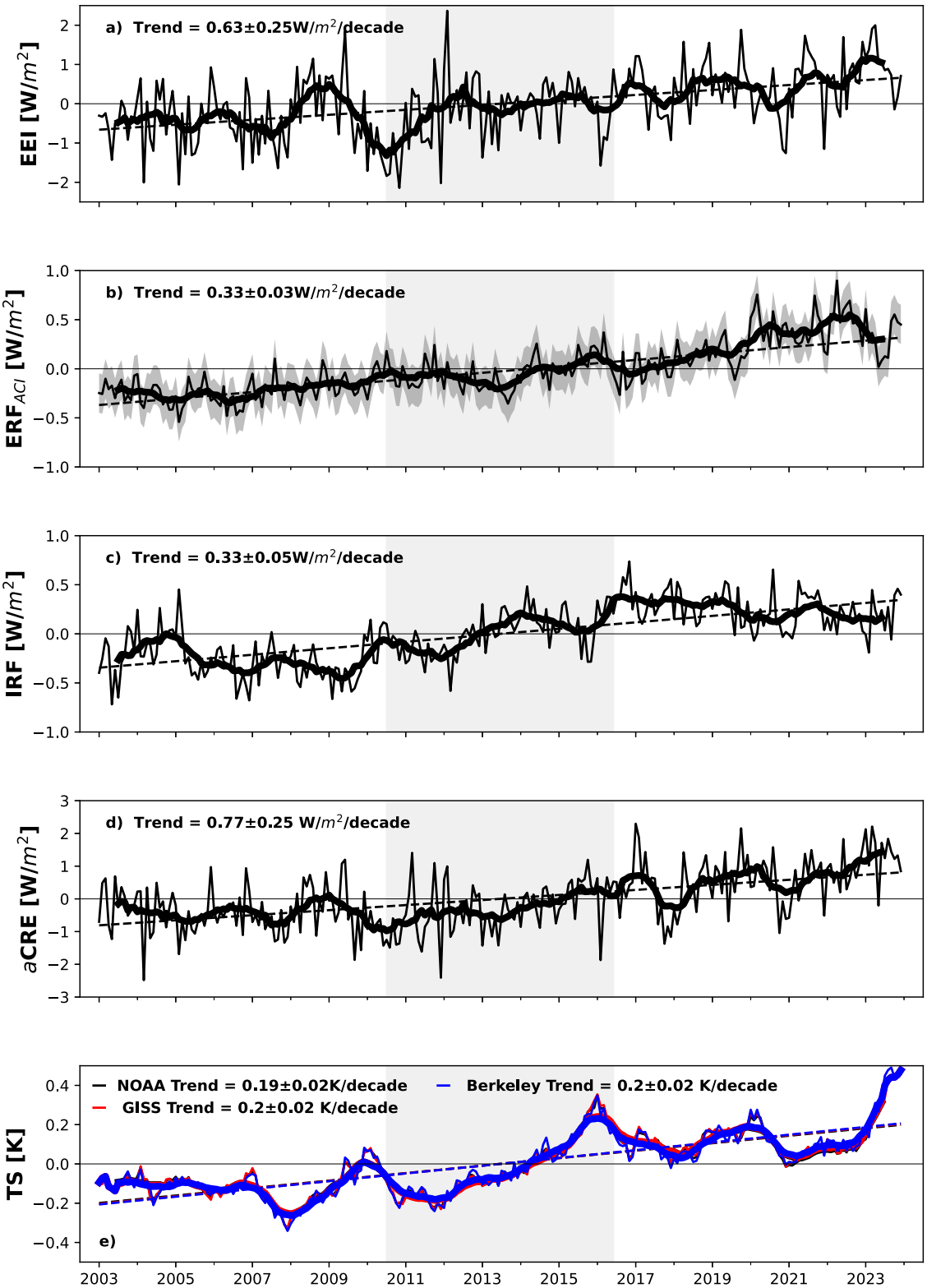


Fig. 2 | Time series of variables of interest. CERES Earth's Energy Imbalance (EEI) (**a**), Effective radiative forcing from aerosol-cloud interactions, ERF_{ACI} (**b**), combined instantaneous radiative forcing (IRF) of aerosol direct effect and greenhouse gases (**c**), adjusted cloud radiative effect (aCRE), which is the sum of ERF_{ACI}, shortwave cloud feedback, and cloud masking effect (see "Methods"), (**d**), and global mean temperature (**e**) for the last two decades (2003–2023) over the oceans between 60°S and 60°N. The mean of each variable in (**a**–**d**) is removed from time series first. The linear trend of each variable is also calculated and shown for each panel. The thick lines represent 12-month running mean while the thin lines are the original data.

Table 1 | SW ERF_{ACI} trends over the globe or oceans between 60°S and 60°N from different studies that employed different sets of models and analysis methods

Study	Models/methods	SW ERF_{ACI} trend (global)	SW ERF_{ACI} trend (60°S–60°N Ocean)
Raghuraman et al.[15]	RFMIP models	0.06	/
Hodnebrog et al.[27]	4 models	0.15[a]	/
Park and Soden[31]	5 RFMIP models	/	0.14
	Regression-based	/	−0.004 to −0.006
This study, assuming zero forcing over land	Cloud adjustments	0.21	0.33
This study, including Twomey effect over land	Cloud adjustments	0.25	0.33

Trends from these different methods and models differ strongly.
[a]From Figure S5b of that study.

cloud adjustments to greenhouse gas forcing[3,13,15]. During the post El Nino sub-period, ERF_{ACI} is the leading forcing component (Fig. 2b) contributing to the EEI increase (Fig. 2a) as instantaneous greenhouse gas forcing shows almost a flat trend[3,13] (Fig. 2c). The trend in aCRE (Fig. 2d) remains stronger than that from the combination of ERF_{ACI}, forcing from greenhouse gases and ARI (Fig. 2c), suggesting strong contribution from cloud feedback. The strong increase in EEI (Fig. 2a) since 2011 appears to be related to strong ERF_{ACI} and shortwave cloud feedback[19,25,27,36].

The spatial distributions of trends in aCRE, ERF_{ACI}, their difference, and SW instantaneous forcing ARI are shown in Fig. 3. Trends in aCRE are about five times stronger than those of ERF_{ACI} regionally, and also have much greater spatial variability (standard deviation of 1.65 Wm^{-2}/decade compared to 0.33 Wm^{-2}/decade for ERF_{ACI}), with compensating large positive and negative values[19,25,36]. Subtracting the contribution of ERF_{ACI} from aCRE gives us the sum of temperature mediated cloud radiative response and cloud adjustments to greenhouse forcing (see "Methods"), which changes only slightly the apparent geographical distribution (like for the large blob of increasing trend in the North Pacific and the small conspicuous blob off the coast of California and Baja California)[37]. The increasing trend over the Gulf stream off the east coast of North America is reduced due to strong contribution of ERF_{ACI} here, and a similar behavior is seen in the oceans west of Western Europe. The increasing trend of aCRE is also substantially reduced by ERF_{ACI} in the south Atlantic basin. On the other hand, the spatial distributions in the tropical western Pacific Ocean and the Indian Ocean are only slightly affected. For comparison, ARI forcing trends are much weaker in magnitude. Again, ARI forcing trends hint at a weak increasing trend in aerosol loading in the remote oceans of SH. The small red blob in Fig. 3d near the Antarctic coast is likely an artifact of the linear assumption adopted radiative kernel method. Specifically, applying a constant surface albedo kernel, developed under certain base conditions, to sea-ice areas undergoing melting or cloud thickening leads to inaccurate estimates (see Methods section for more detailed discussion).

Removing ERF_{ACI} from the aCRE substantially reduces the regression slope between aCRE and global ocean mean temperature (Fig. 2), an approximate measure of SW cloud feedback (see "Methods" and equation 1). To quantify this, we follow a similar procedure in ref. 15 and regress global ocean aCRE against the mean temperature with and without removing the ERF_{ACI} contribution to the mean. The regression slope changes from 1.46 ± 0.16 Wm^{-2}/K to 0.92 ± 0.16 Wm^{-2}/K, or a decrease of 37%. This highlights the importance of removing ERF_{ACI} from SW aCRE to obtain estimates of cloud feedback[15] even though locally the amplitude of ERF_{ACI} is small (Fig. 3). The strong local variations of aCREs tend to compensate each other, and their global mean contribution is comparable to the ERF_{ACI} (Figs. 2b and 3).

Zonal mean trends for various quantities are shown in Fig. 4. Both EEI and absorbed solar radiation have a pronounced peak at around 40°N. A peak is seen at similar latitude for zonal mean trends of both N_d and ERF_{ACI} as well as mean temperature. The peak in N_d is likely a result of reduced anthropogenic emission in the NH mid-latitudes. This mid-latitude peak is also observed for combined instantaneous forcing of greenhouse gases and ARI and SW aCRE. Some secondary peaks are seen in the SH subtropics, more so for variables such as ASR, aCRE, and ERF_{ACI}. In the SH, zonal mean trends exhibit a weaker peak around 10°S for absorbed solar radiation, EEI, ERF_{ACI} and N_d.

Hemispheric contrast

During the last two decades, the decreasing trend of N_d is much stronger in the NH than the SH (Fig. 1). The NH trend is mostly driven by decreasing anthropogenic emissions, especially sulfates from East Asia[29], North America, and Europe, while increasing emissions from India and southeast Asia counteract the decrease only weakly. The trend of SH may be driven by an increase in natural emissions, as suggested by positive trends of aerosol loading and N_d over remote oceans, and a negative trend in parts of coastal regions (Figs. 1 and S2). This hemispheric asymmetry in N_d anomalies, and resulting ERF_{ACI} and forcing from ARI can contribute to hemispheric contrast in EEI[12], which can affect large-scale circulation and precipitation patterns[38–40].

Here, we analyze the contribution of ERF_{ACI} and other forcings to the oceanic hemispheric EEI contrast. Fig. 5a shows the time series of hemispheric EEI contrast - SH mean minus NH oceanic mean between 60°S and 60°N. The contrast exhibits a −0.57 ± 0.26 Wm^{-2}/decade trend, a result of EEI of the NH oceans increasing at a higher rate than that of the SH oceans. The hemispheric contrast in ocean EEI can be decomposed into its two components, absorbed solar radiation, −0.52 ± 0.36 Wm^{-2}/decade (Fig. 5b), and outgoing LW radiation, 0.05 ± 0.31 Wm^{-2}/decade (not shown). Absorbed solar radiation is thus the leading contributor. We further decompose the absorbed solar radiation into clear sky and cloudy parts. The trend of the hemispheric contrast of SW aCRE is strong at −0.37 ± 0.34 Wm^{-2}/decade, while the trend of the hemispheric contrast of ERF_{ACI} is −0.18 ± 0.06 Wm^{-2}/decade, accounting for about half the the aCRE contrast. Hemispheric contrast in forcing from ARI is also substantial at −0.19 ± 0.05 Wm^{-2}/decade[12,13] though the global mean trend of forcing from ARI is small, reflecting the strong hemispheric asymmetry in aerosol emission changes (see Fig. 4). Shortwave forcing from ARI is heavily concentrated in the NH while both ERF_{ACI} and shortwave aCRE are rather more evenly distributed between the hemispheres, which points to an important role of ARI in driving the SH-NH contrast in EEI and ASR[15,41,42]. Overall, total aerosol forcing, including both ARI and ERF_{ACI}, has contributed 71% to the trend in hemispheric contrast of ASR over oceans.

The observed full hemispheric contrast in EEI has a small and statistically insignificant trend of −0.13 ± 0.23 Wm^{-2}/decade. The large hemispheric contrast in EEI over oceans translates to −0.23 ± 0.16 Wm^{-2}/decade in effective hemispheric contrast (see Methods), which is offset by the high-latitude, i.e., 60°–90° in both SH

https://doi.org/10.1038/s41467-026-72926-3

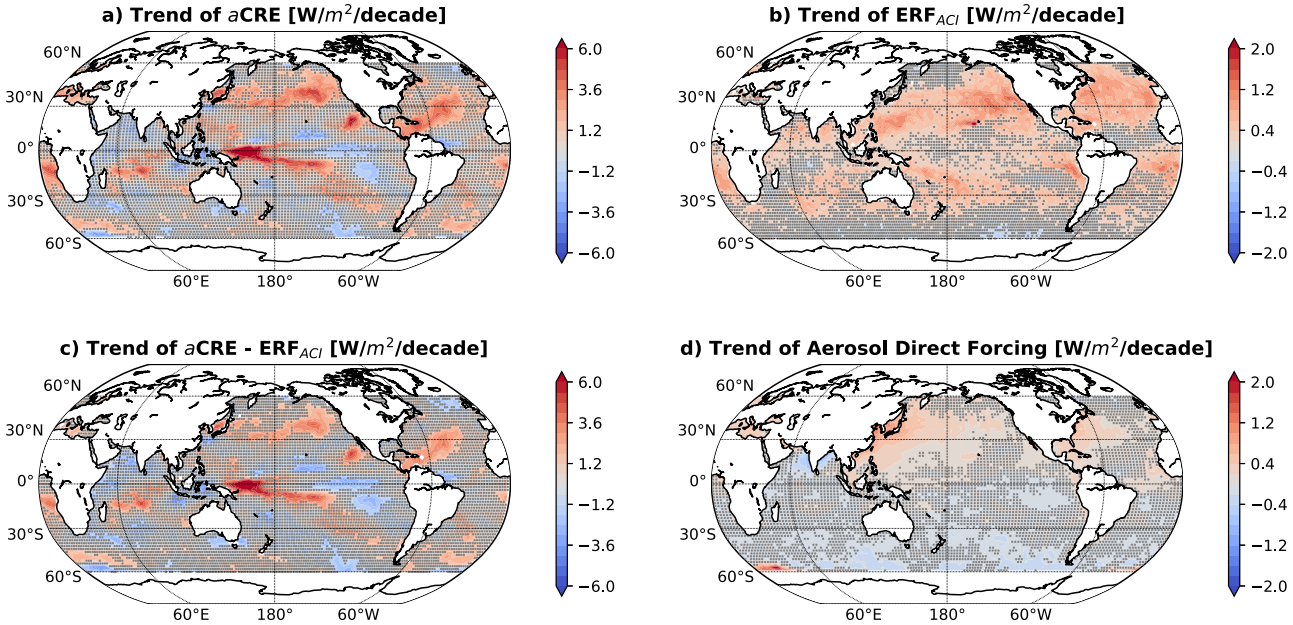


Fig. 3 | . Trend maps of adjusted cloud radiative effect (aCRE) (**a**), effective radiative forcing from aerosol-cloud interactions (ERFACI) (**b**), aCRE minus ERFACI (**c**), and aerosol direct forcing from aerosol-radiation interactions (**d**). Note the difference in color bar scales between (**a**–**d**). Similar to Fig. 1, stippled areas are for $p > 0.05$. Removing ERF_{ACI} from aCRE produces small but noticeable changes as shown (**c**).

and NH, effective hemispheric contrast of +0.11 ± 0.08 Wm^{-2}/decade, resulting in a near flat trend for the full hemispheric contrast in EEI. Absorbed solar radiation is the main driver of the global hemispheric contrast with a trend −0.25 ± 0.28 Wm^{-2}/decade, to which the combined instantaneous aerosol direct forcing and ERF$_{ACI}$ over the oceans contribute −0.15 ± 0.22 Wm^{-2}/decade, or 60%. Total aerosol forcing is also a major contributor to the full hemispheric contrast in absorbed solar radiation and EEI, and the near symmetry in global hemispheric contrast of EEI is a result of compensating impacts from different processes.

Discussion and conclusions

The fact that ERF$_{ACI}$ can rival instantaneous greenhouse gas forcing over decadal timescales according to our analysis has important implications for understanding the past and predicting future decadal climate changes. The decline in global mean temperature from the 1940s to the 1970s coincided with a steep rise in anthropogenic sulfur emissions (Fig. S2), whereas the accelerated warming since the 1980s aligns with a reversal of this trend (Figs. S1 and S6 and refs. 43,44). Given the accelerated increase in emissions between the 1940s and 1970s, the resulting ERF$_{ACI}$ change was likely stronger than in the past two decades—potentially enough to temporarily overpower greenhouse gas forcing and cause net cooling. The warming effect of ERF$_{ACI}$ (Fig. 2) during the last decade contributes to the EEI increase, which conspires with other factors in driving the rapid warming rate in recent years (Fig. 2,[19,25,33,36]). Looking ahead, the ongoing and projected future decline in aerosol emissions, coupled with rising greenhouse gas forcing, is expected to amplify near-future warming. Moreover, the 'pattern effect' of cloud feedback may have dampened warming since the 1980s; if this effect weakens or reverses in the future, it could further increase the future warming rate[45–47]

The reduction in anthropogenic aerosol emissions in the past two decades is effectively an inadvertent global scale reverse geoengineering event[19], which has contributed substantial warming effect based on our estimates and whose cost must be considered. While the shift to cleaner energy sources and strict air quality measures benefit human health, it also reduces aerosol loading which exacerbates the rate of warming. For example, the shipping fuel regulations of 2020 have likely introduced additional warming[19,25] with a range of possible adverse impacts. Should the costs of rapid warming prove substantial, research and evaluation of short-term solar radiation modification strategies such as marine cloud brightening need to be better understood[48] as a companion option to the long-term goal of reduced greenhouse gas emissions. The strong ERF$_{ACI}$ trend in this study indicates that aerosol forcing is close to the high end of model ensemble estimates and solar radiation modification strategies may be effective at temporarily reducing warming rate even though they have other consequences.

It is important to note that estimating ERF$_{ACI}$ remains challenging due to various sources of uncertainty[5]. For example, MODIS estimates of N_d can be biased under certain observational conditions. When exploring the sensitivity of our findings to the details of filtering MODIS data at the pixel level[49] we found that application of filters tends to produce stronger negative trends in N_d and therefore higher ERF$_{ACI}$ estimates (Figs. S3 and S4). Different approaches of estimating ERF$_{ACI}$ can also produce different results. For example, multi-model mean of ERF$_{ACI}$ suggests a weaker trend at around 0.06 ± 0.06 Wm^{-2}/decade between 2001 and 2020[15] while other observation-based estimates suggest even stronger ERF$_{ACI}$ than ours[17,23]. Recent studies also show diverging results regarding the magnitude of ERF$_{ACI}$'s contribution to the recent EEI increase, from near zero[31] to substantial (like our estimate)[50], suggesting that ERF$_{ACI}$ is still uncertain depending on the methods and models (see Table 1). The small uncertainty of the trend reported here thus should not be interpreted as the uncertainty for ERF$_{ACI}$ is reduced. Instead, this only reflects the uncertainty in our own method that has several assumptions that may reduce the uncertainty range (see "Method"). More work is needed to take advantage of the observational constraints in the last two decades and further improve ERF$_{ACI}$ characterization. There is a need for reliable ERF$_{ACI}$ estimates over land trends to better calculate their true global impact.

The strong ERF$_{ACI}$ and its hemispheric asymmetry also have implications for understanding recent shifts in warming pattern in the Pacific. Theory and modeling work suggest that a strong subtropical to mid-latitude aerosol forcing in the NH can induce a fast cooling of sea

https://doi.org/10.1038/s41467-026-72926-3

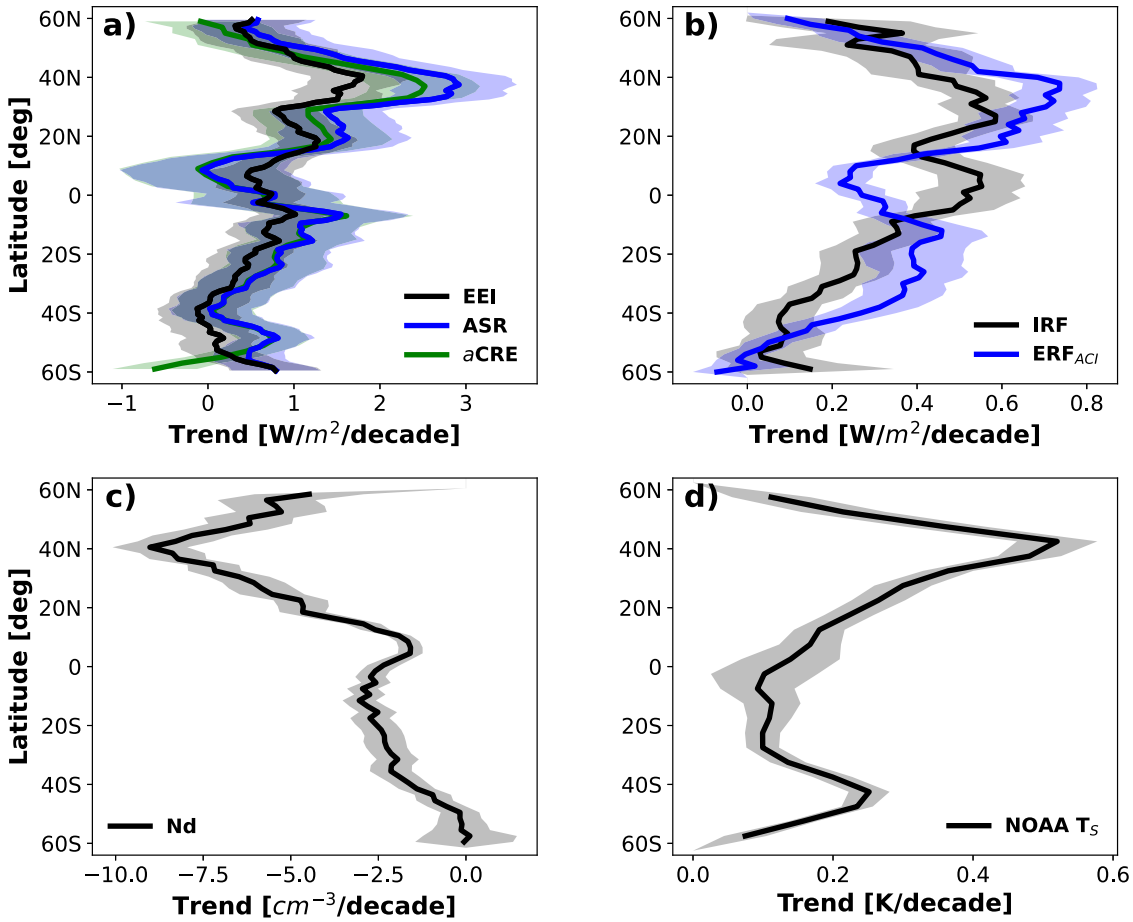


Fig. 4 | Zonal mean of trends of different variables. Earth's energy imbalance (EEI), absorbed solar radiation (ASR), and adjusted cloud radiative effect (aCRE) (**a**), combined forcing from greenhouse gases and aerosol direct effect (IRF) and effective radiative forcing from aerosol-cloud interactions (ERF_{ACI}) (**b**), N_d (**c**), and mean temperature T_s from the National Oceanic and Atmospheric Administration (**d**). The shading for each variable represents the 95% confidence interval range (see "Methods").

surface temperature in the Eastern Pacific followed by a warming trend in the next 10–15 years[51–53]. The increase of ERF_{ACI} and aerosol-induced hemispheric contrast in the last 10–15 years could trigger a warming pattern in the tropical Pacific with far-reaching global impacts. Aerosol-related forcing is a major contributor to the observed trend in hemispheric contrast of absorbed radiation and EEI, which may in turn affect precipitation patterns[40,41,54,55].

The ERF_{ACI} trend reported here, at around 0.21 Wm^{-2}/decade when scaled up to a global mean, is much stronger than the multi-model mean estimate of around $0.06 Wm^{-2}/decade$[15]. Our estimate reduces the 0.2 Wm^{-2}/decade gap between modeled and observed EEI trend of the CERES record in the SW[27]. It also suggests a muted total SW cloud feedback during this period because a stronger ERF_{ACI} would suggest less contribution from cloud feedback to the aCRE[15]. We show that it is important to account for the contribution of ERF_{ACI} when studying cloud feedback because of its persistence in sign, despite the smaller magnitudes compared to aCRE at the local level.

Finally, our study is a good example of why availability of long-term, inter-calibrated, and stable spaceborne observations, whose future is not ensured, is invaluable for both scientific research and informed policy making[3,33,56]. Such datasets provide the crucial observational anchor needed to unravel uncertain processes such as aerosol-cloud interactions and cloud feedback. While this work adds to the growing body of literature that underscores the value of records of sustained observational continuity, many such datasets are nevertheless at risk of discontinuation.

Methods

Data

We use both level-2 and level-3 Aqua MODIS data[21]. Retrievals of cloud optical depth and cloud droplet effective radius are performed at the pixel level and then aggregated into gridded level-3 data. N_d and Liquid water path are derived from these two retrievals. For N_d, we adopt the approach of ref. 49. MODIS retrievals have known biases and uncertainties associated with various factors and several approaches have been proposed to filter level-2 data to mitigate these issues. We tested these approaches and assessed their impact on derived trends (see Figs. S3 and S4). In the main text, we report results using equation 11 from ref. 49 to derive N_d from the original level-3 data due to two reasons. First, they do not exclude any data points that have valid retrievals, where retrieval biases can cancel each other, while filtering data can introduce sampling biases. Second, using the same data selection, ERF_{ACI} calculations from our previous studies show consistent results with independent methods and modeling work[19,24,25]. We use the CERES energy balanced and filled (EBAF) Ed. 4.2 products. They include the shortwave reflected fluxes, outgoing longwave radiation fluxes, both shortwave and longwave CREs, and cloud albedo. EEI is derived from EBAF variables[10]. The NASA's Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA2) reanalysis data are used to separate cumulus clouds from stratocumulus cloud regimes when testing the impact of applying cloud adjustments derived from ship-tracks to the cumulus regime.

https://doi.org/10.1038/s41467-026-72926-3

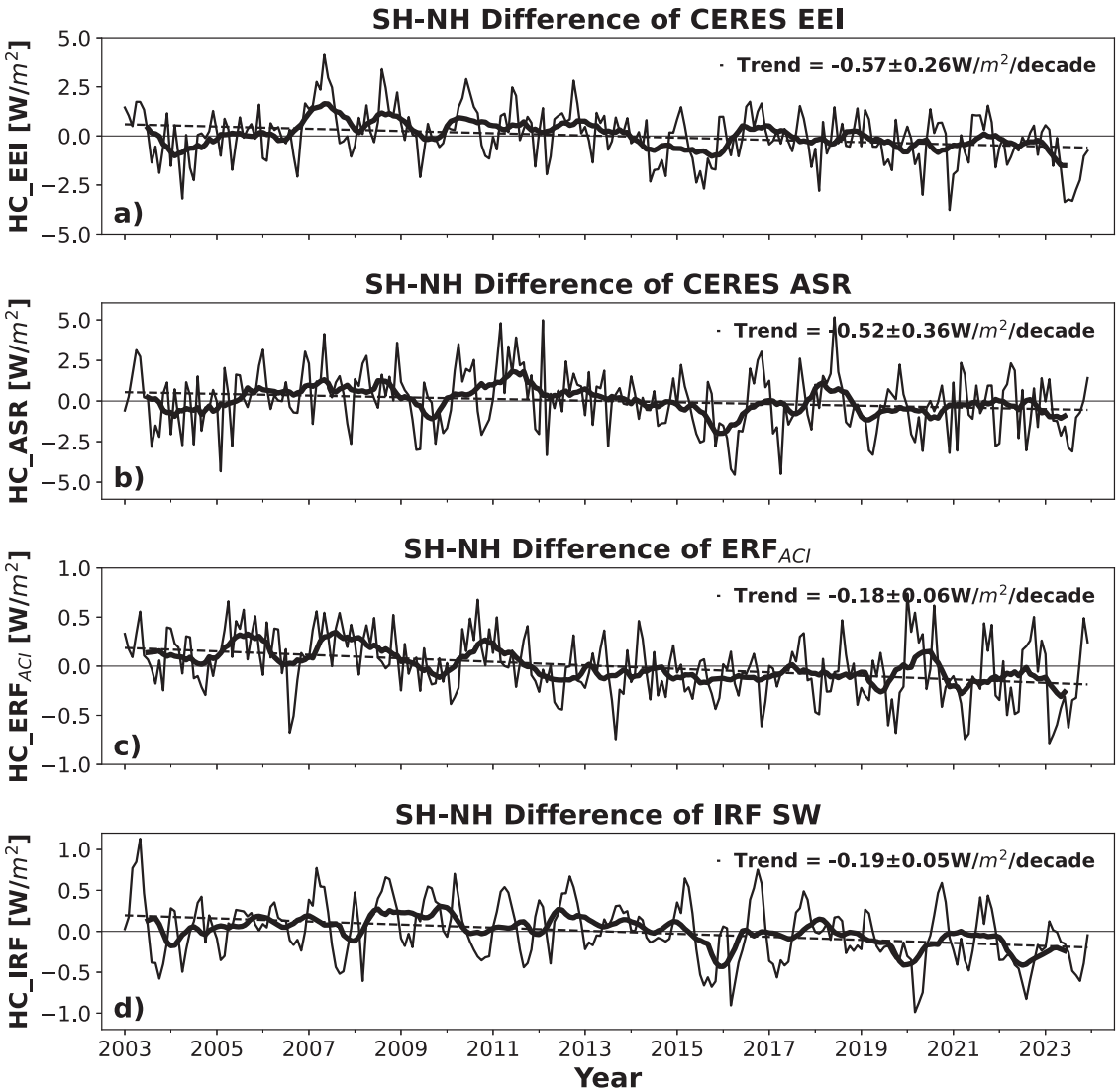


Fig. 5 | . Time series of hemispheric contrasts, southern hemisphere (SH) minus northern hemisphere (NH), in different variables, namely CERES Earth's energy imbalance (EEI) (**a**), CERES absorbed solar radiation (**b**), effective radiative forcing from aerosol-cloud interactions (ERF_{ACI}) (**c**), and shortwave (SW) instantaneous radiative forcing from aerosol radiation interaction (**d**). The thin lines represent the raw time series and the thick lines in each panel represent the 12-month running mean.

Global mean temperature records from the NOAA National Centers for Environmental Information[58], Goddard Institute of Space Studies[59], and Berkeley Earth[60] are used.

Methods

We combine observation-based radiative kernel calculations of forcing from greenhouse gases and ARI with observationally estimated ERF_{ACI} to quantify contributions of different drivers of the strong changes in EEI during the CERES period (see *Definition of terms and their relationships*). We apply cloud adjustments derived from ship-tracks to monthly gridded satellite data to estimate ERF_{ACI}[16,19,23,61], following similar formulations as in previous studies[16,23,62]. Technical details of how cloud adjustment functions are derived and ERF_{ACI} calculation performed are discussed in *Cloud adjustment functions* and *Calculating* ERF_{ACI} subsections, respectively.

Definition of terms and their relationships. In this study, we have

$$EEI = IRF + \Delta R_\lambda + Adj_{str} \quad (2)$$

where IRF is the instantaneous radiative forcing, ΔR_λ includes adjustments and surface temperature-mediated feedbacks, and Adj_{str} is the stratospheric adjustments to GHG forcings[13]. ΔR_λ can be decomposed into:

$$\Delta R_\lambda = \Delta R_{non-clouds} + \Delta R_C \quad (3)$$

where $\Delta R_{non-clouds}$ and ΔR_C are the radiative feedback and adjustments from non-cloud variables and clouds, respectively.

Cloud radiative effect (CRE) is the difference between all-sky and clear-sky radiative fluxes and can be determined by CERES. While this includes radiative responses to cloud changes, it also includes the difference in the radiative effects of non-cloud perturbations between all-sky and clear-sky conditions[63]. To isolate radiative changes solely due to clouds, we calculate the change in CRE minus these so-called "cloud masking" effects (CM). As in ref. 13, we consider this the adjusted cloud radiative effect (aCRE) and compute it using CRE from CERES-EBAF Ed 4.2 and cloud masking terms diagnosed from applying all-sky and clear-sky radiative kernels as described above.

By this definition, aCRE includes ERF_{ACI}, cloud radiative response from temperature-mediated changes (CRR), and cloud adjustments to greenhouse gas forcing ($CloudAdj_{GHG}$)[13,15], or

$$aCRE = CRE - CM = ERF_{ACI} + CRR + ERF_{GHGSW} + \epsilon \quad (4)$$

where CRE is cloud radiative effect from CERES observations, CM is cloud masking effect that can be calculated[13,15], CRR is the temperature mediated cloud radiative response, i.e., the true cloud feedback term. ERF_{GHGSW} are ERF due to cloud adjustments to greenhouse gas forcing in the shortwave. ϵ represents the residual and natural variability. Here we only consider the SW part of aCRE since the LW part of the EEI trend can be well accounted for ref. 13. From equation 3, it is clear that if ERF_{ACI} contributes meaningfully to *aCRE*, estimating of cloud feedback based on *aCRE* alone will introduce error[15].

The EEI can be simply written as

$$\Delta EEI = \Delta ASR - \Delta OLR \quad (5)$$

using CERES observations of net solar fluxes and outgoing longwave radiation. Multiple processes contribute to ΔASR such as changes in ΔCRE, ERF_{ACI}, surface albedo, aerosol direct forcing.

Cloud adjustment functions. Cloud adjustment functions are based on numerous ship-track samples[23]. We automatically find background cloud pixels for each ship-track segment and differentiate cloud properties between ship-track and background pixels. For cloud fraction difference, overlap between high and low clouds on the true low cloud fraction is considered[23]. Only low cloud pixels are included in calculating cloud properties. N_d is calculated using cloud retrievals of cloud droplet effective radius, r_e, and cloud optical depth, τ_c, with equation 11 in ref. 49. For each segment, we calculate cloud changes in LWP, N_d, cloud fraction, and r_e etc. using the means of ship-track and background pixels. The cloud adjustments are computed as the ratio of changes in cloud fraction and LWP over changes in N_d.

Distributions of ΔLWP are known to be not normal and have strong outliers and ΔN_d can be small in some of the ship-track samples. When calculating $\frac{dln\,\mathrm{LWP}}{dln\,N_d}$, the combination of large ΔLWP and small ΔN_d can introduce outlier values. We remove the top 5% absolute values from calculated $\frac{dln\,\mathrm{LWP}}{dln\,N_d}$ or use kernel density estimation method to remove outliers before calculating the mean $\frac{dln\,\mathrm{LWP}}{dln\,N_d}$.

We then bin cloud adjustments based on different variables such as control cloud N_d, above cloud humidity, and background cloud r_e, with which we can obtain cloud adjustments as a function of these variables. In our study, we use the cloud adjustment function form that gives the most conservative estimate of ERF_{ACI}[23] that is also most consistent with independent estimates. It is important to note that our calculated ERF_{ACI}[23] does not fully account for the atmospheric and non-cloud adjustments to it[6].

The derived cloud adjustments are robust when we select samples from different periods or randomly select 50% of the samples (see Supplementary Information).

Calculating ERF_{ACI}. First, we consider the simple case of ERF_{ACI} without considering aerosol effects on cloud fractions. Cloud albedo (A_c) sensitivity to aerosols can be taken as the sum of the Twomey effect and aerosol induced LWP adjustments[16,23]:

$$S = \frac{dA_c}{dN_d} = \frac{A_c(1-A_c)}{3N_d} \times \left(1 + \frac{5}{2}\frac{dln\,LWP}{dln\,N_d}\right) \quad (6)$$

where S is the susceptibility of cloud albedo to droplet number concentration. We have

$$\Delta SW_{TOA} \approx -SW_{downwelling} \times f_c \times S \times \Delta N_d \quad (7)$$

ERF_{ACI} from the *T*womey effect and LWP adjustment is therefore:

$$ERF_{ACI,\,Twomey\&LWP} \approx -SW_{\mathrm{downwelling}} \times f_c \times A_c \times (1-A_c) \times \left(\frac{1}{3} + \frac{5}{6}\frac{dln\,LWP}{dln\,N_d}\right) \times \Delta lnN_d \quad (8)$$

$SW_{downwelling}$ is shortwave flux at the cloud level. It can be conservatively estimated using the CERES clear sky surface downwelling shortwave flux and we approximate changes in SW at TOA with above cloud changes. f_c is low cloud fraction from MODIS level-3 product. A_c is the cloud albedo, it can be calculated from the CERES all-sky albedo (A_{allsky}), clear-sky albedo (A_{clearsky}) and total cloud fraction (f_{total}) using the equation

$$A_c = (A_{\mathrm{allsky}} - (1 - f_{\mathrm{total}}) \times A_{\mathrm{clearsky}})/f_{\mathrm{total}}. \quad (9)$$

for all ship-track samples and sort them by their control cloud N_d, and calculated mean values of $\frac{dln\mathrm{LWP}}{dlnN_d}$ for each control N_d bin. The dependence of $\frac{dln\mathrm{LWP}}{dlnN_d}$ on N_d is used as our function to calculate the LWP adjustment. We first calculated 2003–2023 climatological-mean N_d from the MYD08_M3 data. Based on the 1-D dependence of $\frac{dln\mathrm{LWP}}{dlnN_d}$ on N_d we derive the global gridded LWP adjustment sensitivity.

Aerosols can also affect cloud fraction. To consider the effect of Cf adjustment, we consider the sensitivity of scene albedo ($A = A_{ac}f_{\mathrm{total}} + A_s(1 - f_{\mathrm{total}})$) to N_d, we have:

$$S^* = \frac{dA}{dN_d} = \frac{d(A_{\mathrm{ac}}f_{\mathrm{total}} + A_s(1-f_{\mathrm{total}}))}{dN_d} \approx f_c \times S + \left(1 - f_{\mathrm{high}}\right) \times \frac{df_c}{dN_d} \times (A_{ac} - A_s) \quad (10)$$

where A is the all-sky scene albedo, i.e., including both cloudy, A_{ac}, and clear, A_s, parts; f_{total} and f_c are all cloud and low cloud fraction obtained from the MYD08_M3 data; A_s is the surface albedo, derived from the CERES EBAF-TOA data; $1 - f_{\mathrm{high}}$ is used to take into account of effect of overlap on f_c adjustment and derived from the MODIS cloud product. Here, we simplified the calculation by approximating cloudy albedo at the top of the atmosphere by that at the cloud top and clear sky albedo by surface albedo since the atmospheric effect on the cloudy and clear sky albedos partially cancels. We assume zero aerosol effects on high cloud fraction. $\frac{df_c}{dN_d}$ is calculated using our ship-track samples, similar to the LWP adjustment. We account for the cloud overlap between high and low clouds. For more details, we refer to our previous studies[19,23].

Once we derive S^* with equation 10, we can get changes in TOA flux by

$$ERF_{\mathrm{ACI}} = -SW_{\mathrm{downwelling}} \times S^* \times \Delta N_d \quad (11)$$

More detailed discussions about sensitivity and uncertainty of our approach can be found in our previous studies[19,23].

We obtain N_d changes for the last two decades using MODIS retrievals. We derive N_d based on MODIS retrievals at monthly time scales at a $1^o X 1^o$ resolution. For each grid, we remove the climatological mean and the seasonal cycle to derive ΔN_d time series. Using the equations above and the derived ΔN_d, we can calculate ERF_{ACI} for each grid point at monthly time scales. Alternatively, we tested the

sensitivity of derived ΔN_d to different filters of the MODIS retrievals and results are reported in the supplementary material[49].

Comparison with independent methods. Our calculated ERF_{ACI}'s have been independently compared to other estimates under different scenarios. For example, our ERF_{ACI} correlates well with CMIP6 multi-model mean estimates between 2003 and 2019 (Fig. S5, $r = 0.89$, $p < 0.01$). We estimate the ERF_{ACI} between pre-industrial and present day to be $-1.1 \pm 0.8\ Wm^{-2}$, which is close to the consensus number of $-0.84 \pm 0.6\ Wm^{-2}$ in the Sixth Assessment Report[6]. It is also very similar to an independent estimate ($-1.02 \pm 0.43\ Wm^{-2}$) also based on MODIS data[17]. We also applied our method to estimate ERF_{aci} from shipping fuel regulation[19] our estimate was very close to values from independent modeling studies at around 0.1–0.15 Wm^{-2} globally[25]. ERF_{ACI} from our method is compared to and consistent with independent estimates based on top-down geospatial krigging approach in the southeast Atlantic shipping lane[19,61]. Both methods show around $+1\ Wm^{-2}$of warming due to reduced ship emissions in the core of the shipping corridor. These agreements indicate that our method provides reasonable estimates of ERF_{ACI} by applying cloud adjustments from ship-tracks to larger scales. The sum of ERF_{ACI} and IRF in our study is $+0.66\ Wm^{-2}$, which is close to $+0.71\ Wm^{-2}$ (2001–2024) using machine learning-based top-down estimates[64] although for slightly different periods.

Such agreements are achieved using cloud adjustments as a function of background N_d and all MODIS cloud retrievals without filtering and represent our best estimate. The reported uncertainty range of ERF_{ACI} trend only includes the uncertainties in the magnitude of cloud adjustment functions and from the MODIS retrievals, but does not consider the uncertainty introduced by forms of cloud adjustments[23] and different filtering methods (see more in next subsections).

Assumptions. We note a few assumptions and simplifications made in our method. The LW effect of aerosol-cloud interactions in low clouds is not considered. This effect is of secondary importance due to the small temperature contrast between low clouds and surface, with a previous study suggesting that it accounts for roughly 10% of the SW ERF_{ACI}[6,17]. The cloud adjustments used here apply best to the stratocumulus regime because most ship-tracks form in this cloud regime, but we assume that they also apply to trade cumulus clouds as well. Comparing results from our method with previous studies suggests that our method estimates smaller cloud fraction changes in response to N_d than other published studies for the trade cumulus clouds[65,66], which would make our estimated ERF_{ACI} for these clouds a lower bound. Cloud adjustments to aerosols can evolve significantly with time. The adjustments derived from ship-tracks represent a time-average behavior which may not be strictly representative because ship-track clouds have different ages[67]. It is also important to note that unlike modeling estimates of ERF_{ACI} that include effects of atmospheric responses to aerosol changes our method only considers cloud responses to microphysical changes. Other assumptions and uncertainties are discussed in ref. 23.

Radiative kernel calculations. Widely used in the decomposition of radiative changes, radiative kernels are a set of pre-computed partial derivatives representing the sensitivity of radiation to a perturbation in a specific state variable. As referred to in the section above, radiative kernels are used in this work to diagnose the non-cloud and cloud radiative feedback and adjustment terms contributing to total ΔR_λ and subsequently to diagnose the aerosol direct forcing as a residual calculation. The non-cloud terms ($\Delta R_{\text{non-clouds}}$) include feedbacks and adjustments in temperature (ΔR_T), water vapor (ΔR_q) and surface albedo (ΔR_a). Following[68] and other work cited herein, each of these non-cloud terms can be diagnosed as the product of a radiative kernel (K^x) and the actual change in variable x:

$$\Delta R_{\text{non-clouds}} = \sum \Delta R_x \approx \sum \left(\frac{\partial R}{\partial x} \times \Delta x \right) \approx \sum (K^x \times \Delta x) \quad (12)$$

where a clear-sky and all-sky K^x allows one to diagnose clear- and all-sky ΔR_x, respectively. While these ΔR_x terms are not directly a focus of this work, the difference between the all-sky and clear-sky terms provide an estimate of the Cloud Masking (CM) effect needed to derive the cloud radiative feedback and adjustment term (ΔR_c), as noted in the section above (Eqs. S8 and S9). The kernel-derived estimates of ΔR_x are also used to diagnose the aerosol direct radiative forcing as a residual calculation. Namely, the clear-sky aerosol direct forcing is estimated as the difference between the clear-sky ΔASR and the sum of the clear-sky, shortwave ΔR_x terms. Then, the all-sky IRF presented in this work is derived by scaling the clear-sky IRF by a constant that accounts for cloud masking[13,68].

Each of these terms can be diagnosed from satellite observations over the CERES period. We extend previous estimates of these terms by ref. 13, similarly applying CloudSat-based radiative kernels[69] to satellite-observed anomalies of surface albedo from CERES-EBAF Ed 4.2, and water vapor and temperature anomalies from AIRS-CLIMCAPS V2[70] to diagnose the non-cloud radiative responses. CERES-EBAF Ed 4.2 is a more recent edition compared to what was used in ref. 13 and CLIMCAPS serves an alternative to the since-discontinued AIRS V6 operational product. As in ref. 13, the clear-sky IRF is diagnosed as the difference between clear-sky net radiative anomalies from CERES and the sum of clear-sky radiative responses. Then, the all-sky IRF presented in this work is derived by scaling the clear-sky IRF by a constant that accounts for cloud masking[13,68]

The small, red blotch in Fig. 3d along the Antarctic Coast directly South of South Africa is likely an artifact of using radiative kernels to diagnose the ARI. By definition, a radiative kernel assumes linearity in the sensitivity of radiation to the perturbation in a given state variable. This holds well for nearly all use cases, but not so well for the application of surface albedo kernels over sea-ice, which has a dependence on the base state. For instance, although the kernel assumes the sensitivity of SW radiation to an albedo change to be a fixed sensitivity, in reality shortwave radiation becomes less sensitive to a given albedo change as the underlying surface warms, ice melts, and local clouds thicken[71]. This can lead to biases in estimates of the surface-albedo radiative feedback when the constant kernel is being applied to an albedo perturbation that occurs under ice and atmospheric conditions different enough from those used to originally develop the kernel. Since the ARI forcing estimate is a residual calculation, any bias in the estimate of the surface-albedo radiative feedback will, in turn, lead to a bias in the residual calculation of the forcing. When applied to observations, as in this work, the bias is localized over a small area and has negligible impact on the global-mean ARI forcing estimate[13].

Calculating confidence interval, uncertainty, and trend analysis. 95% confidence intervals are listed for trends reported in this study. The confidence intervals are calculated as[15] $CI = 1.96X\sqrt{\sigma_{obs}^2 + \sigma_{var}^2}$, where $2\sigma_{obs} = 0.2\ Wm^{-2}$/decade, and σ_{var} is the standard error of the linear fit to the data that account for autocorrelation.

Trends are calculated based on monthly time series and the slope is multiplied by 120 months/decade to get a decadal trend. Seasonal cycles are removed from the time series and autocorrelation is accounted for. We report trends in **bold** if it has a $p < 0.05$. Otherwise, numerical values are in plain fonts. In the case of statistical significance for trends on maps, we control of the false discovery rate[57].

Since our calculations of ERF_{ACI} use both CERES EBAF and N_d data with equation 11, we have $\sigma_{ERF_{ACI}} = ERF_{ACI} \times \sqrt{\frac{\sigma^2_{CERES}}{CERES^2} + \frac{\sigma^2_{N_d}}{\Delta N_d^2}}$, assuming the uncertainties in CERES and N_d are independent. The relative uncertainty of CERES data, such as surface albedo and cloud albedo used in equation 11, is affected by uncertainties in the input data as well as the local CRE[10]. At the grid level, input variables such as cloud optical depth and cloud fraction have relative uncertainty on the order of 15%. The grid level uncertainty of ΔN_d is around 30%[49]. If we assume that the variables from different grids are strongly correlated and thus there is a low degree of freedom in the global mean, the relative uncertainties of global mean N_d and CERES data can be conservatively, i.e., towards the highest uncertainty range, estimated to be 20% and 10%, respectively. The absolute ERF_{ACI} is close to $1\,Wm^{-2}$[5,6]. We thus estimate $\sigma_{ERF_{ACI}} \approx 0.2 Wm^{-2}$. This is our conservative estimate of the ERF_{ACI} uncertainty.

Alternatively, we also carry out two analyses to calculate the uncertainty introduced by MODIS retrieval uncertainties through error propagation[72]. In one, we propagate MODIS relative retrieval uncertainties in droplet effective radius (r_{eff}, 30% and cloud optical depth (τ), 10%, at pixel level[49] to calculated mean ERF_{ACI} at monthly time intervals. We first calculate the effective number of degrees of freedom at each grid point for the retrievals by $N_{eff} = N/s_{free}$, where s_{free} accounts for the spatial autocorrelation in N retrieval samples. We then propagate the retrieval uncertainties in r_{eff} and τ to uncertainty in N_d. The relative uncertainty of N_d is obtained with

$$\sigma_{N_d} \approx \sqrt{\left(\frac{\partial N_d}{\partial r_{eff}}\sigma_{r_{eff}}\right)^2 + \left(\frac{\partial N_d}{\partial \tau}\sigma_{\tau}\right)^2}. \quad (13)$$

Using the uncertainty range of N_d, we carry out Monte Carlo simulations to estimate the uncertainty of ERF_{ACI} by drawing samples from the estimated range of N_d and calculating ERF_{ACI} based on the sampled N_d, from which we can obtain an estimate of standard deviation of ERF_{ACI} at each grid point. We then estimate the uncertainty range of ERF_{ACI} for each month based on the standard deviations of ERF_{ACI} at each grid point. In the other method, we assume a grid level uncertainty of 30% for N_d[49] and calculate the ERF_{ACI} uncertainty range for each grid with the Monte Carlo simulations. We finally estimate the uncertainty range of the trend by making weighted least-squares fits that consider the impact of uncertainty at each time step. The final trends and their uncertainty ranges are similar for both methods. The uncertainties from both methods are much smaller than the conservative estimate in previous paragraph.

We note that this uncertainty and error propagation analyses do not include the uncertainty introduced by different cloud adjustment functions[23]. The cloud fraction adjustment function used here is more conservative and the actual CF adjustment effect could be much stronger. We choose this functional form because it produces results that compare favorably with independent studies. The potential underestimation of LWP adjustment to aerosols due to spatial variability of satellite retrievals[72] is not considered in our LWP adjustments. Directionally, this underestimation would make the positve ERF_{ACI} trend stronger during this period if the real LWP adjustment should be stronger.

It is important to also note that the uncertainty of the trend scales with both the point estimate of the ERF_{ACI} uncertainty range, represented by standard deviation $\sigma_{ERF_{ACI}}$, and $N^{-1.5}$, where N is the number of samples in an evenly spaced time series data. In our study, $N = 252$, which can strongly outweigh the impact of the uncertainty range. In other words, the small magnitude in the uncertainty of the trend should not be understood as the uncertainty range itself being small. For example, when using the uncertainty ranges of the IPCC AR6 ERF_{ACI} time series to calculate the uncertainty of the trend for the same period we find the uncertainty of the trend to be $0.04\,Wm^{-2}$/decade despite the uncertainty range of ERF_{ACI} being almost $1\,Wm^{-2}$.

Scaling results to global mean. The ERF_{ACI} results reported here are for oceans between 60°S and 60°N because cloud adjustment functions are derived based on oceanic clouds. To assess their effective global mean impact, we scale them by the ratio of ocean area between 60°S and 60°N to Earth's surface area, which is 0.63. This effectively assumes ERF_{ACI} trends to be 0 everywhere else. Another option to calculate global mean is to include only the Twomey effect over land and assume cloud adjustments from LWP and CF to be zero since we do not have observations, and then scale the resulting forcing by the area ratio of 60°S and 60°N to the globe. We call the means under this assumption "equivalent global means".

For hemispheric contrast in a variable, we similarly scale a hemispheric value by the ratio between the ocean area between 0° and 60° in this hemisphere by half of Earth's surface area. This captures the contribution from ocean areas to that hemisphere. The hemispheric contrast of such scaled values therefore represents the contribution of ocean areas to the full hemispheric contrast. We report these values as "effective" hemispheric contrasts.

Data availability

MODIS data are freely available at https://www.earthdata.nasa.gov/centers/laads-daac. MERRA-2 reanalysis data are also freely availalbe at https://gmao.gsfc.nasa.gov/reanalysis/merra-2/. Radiative kernels and related data are available at https://doi.org/10.5281/zenodo.17825264[73]. CERES data are available at https://ceres.larc.nasa.gov/data/#ebaftoa-level-3. The cloud adjustments and ship-track related data are published in our previous papers[19,23].

References

1. Hansen, J. et al. Earth's energy imbalance: confirmation and implications. *Science* **308**, 1431–1435 (2005).
2. Trenberth, K. E. & Cheng, L. A perspective on climate change from Earth's energy imbalance. *Environ. Res. Clim.* **1**, 013001 (2022).
3. Loeb, N. G. et al. Observational assessment of changes in Earth's energy imbalance since 2000. *Surv. Geophys.* **45**, 1757–1783 (2024).
4. Von Schuckmann, K. et al. An imperative to monitor Earth's energy imbalance. *Nat. Clim. Change* **6**, 138–144 (2016).
5. Bellouin, N. et al. Bounding global aerosol radiative forcing of climate change. *Rev. Geophys.* **58**, 2019–000660 (2020).
6. Forster, P. et al. The Earth's energy budget, climate feedbacks, and climate sensitivity. In *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* 923–1054 (Cambridge University Press,2021).
7. Raghuraman, S. P., Paynter, D. & Ramaswamy, V. Anthropogenic forcing and response yield observed positive trend in Earth's energy imbalance. *Nat. Commun.* **12**, 4577 (2021).
8. Cheng, L. et al. Ocean heat content in 2023. *Nat. Rev. Earth Environ.* **5**, 232–234 (2024).
9. Schuckmann, K. et al. Heat stored in the Earth system 1960-2020: where does the energy go? *Earth Syst. Sci. Data* **15**, 1675–1709 (2023).
10. Loeb, N. G. et al. Clouds and the Earth's Radiant Energy System (CERES) Energy Balanced and Filled (EBAF) Top-of-Atmosphere (TOA) Edition-4.0 Data Product. *J. Clim.* **31**, 895–918 (2018).
11. Loeb, N. G. et al. Satellite and ocean data reveal marked increase in Earth's heating rate. *Geophys. Res. Lett.* **48**, 2021–093047 (2021).
12. Loeb, N. G. et al. Emerging hemispheric asymmetry of Earth's radiation. *Proc. Natl. Acad. Sci. USA* **122**, 2511595122 (2025).
13. Kramer, R. J. et al. Observational evidence of increasing global radiative forcing. *Geophys. Res. Lett.* **48**, 2020–091585 (2021).

14. Stevens, B. Atmospheric moist convection. *Annu. Rev. Earth Planet Sci.* **31**, 605–643 (2005).
15. Raghuraman, S. P., Paynter, D., Menzel, R. & Ramaswamy, V. Forcing, cloud feedbacks, cloud masking, and internal variability in the cloud radiative effect satellite record. *J. Clim.* **36**, 4151–4167 (2023).
16. Toll, V., Christensen, M., Quaas, J. & Bellouin, N. Weak average liquid-cloud-water response to anthropogenic aerosols. *Nature* **572**, 51–55 (2019).
17. Wall, C. J., Storelvmo, T. & Possner, A. Global observations of aerosol indirect effects from marine liquid clouds. *Atmos. Chem. Phys.* **23**, 13125–13141 (2023).
18. Yuan, T., Oreopoulos, L., Platnick, S. E. & Meyer, K. Observations of local positive low cloud feedback patterns and their role in internal variability and climate sensitivity. *Geophys. Res. Lett.* **45**, 4438–4445 (2018).
19. Yuan, T. Abrupt reduction in shipping emission as an inadvertent geoengineering termination shock produces substantial radiative warming. *Commun. Earth Environ.* **5**, 1–8 (2024).
20. Chen, Y. et al. Machine learning reveals climate forcing from aerosols is dominated by increased cloud cover. *Nat. Geosci.* **15**, 609–614 (2022).
21. Platnick, S. et al. The MODIS cloud optical and microphysical products: collection 6 updates and examples from terra and aqua. *IEEE Trans. Geosci. Remote Sens.* **55**, 502–525 (2017).
22. Christensen, M. W. et al. Opportunistic experiments to constrain aerosol effective radiative forcing. *Atmos. Chem. Phys.* **22**, 641–674 (2022).
23. Yuan, T. et al. Observational evidence of strong forcing from aerosol effect on low cloud coverage. *Sci. Adv.* **9**, 7716 (2023).
24. Diamond, M. S. Detection of large-scale cloud microphysical changes within a major shipping corridor after implementation of the International Maritime Organization 2020 fuel sulfur regulations. *Atmos. Chem. Phys.* **23**, 8259–8269 (2023).
25. Gettelman, A. et al. Has reducing ship emissions brought forward global warming? *Geophys. Res. Lett.* **51**, 2024–109077 (2024).
26. Xiong, X. et al. Aqua MODIS: 20 years of on-orbit calibration and performance. *J. Appl. Remote Sens.* **17**, 037501 (2023).
27. Hodnebrog, Ø et al. Recent reductions in aerosol emissions have increased Earth's energy imbalance. *Commun. Earth Environ.* **5**, 1–9 (2024).
28. Quaas, J. et al. Robust evidence for reversal of the trend in aerosol effective climate forcing. *Atmos. Chem. Phys.* **22**, 12221–12239 (2022).
29. Samset, B. H. et al. East Asian aerosol cleanup has likely contributed to the recent acceleration in global warming. *Commun. Earth Environ.* **6**, 543 (2025).
30. Wood, R. et al. The VAMOS Ocean-Cloud-Atmosphere-Land Study Regional Experiment (VOCALS-REx): goals, platforms, and field operations. *Atmos. Chem. Phys.* **11**, 627–654 (2011).
31. Park, C. & Soden, B. J. Negligible contribution from aerosols to recent trends in Earth's energy imbalance. *Sci. Adv.* **11**, 9429 (2025).
32. Smith, C. J. et al. Effective radiative forcing and adjustments in CMIP6 models. *Atmos. Chem. Phys.* **20**, 9591–9618 (2020).
33. Schmidt, G.A. et al. CERESMIP: a climate modeling protocol to investigate recent trends in the Earth's Energy Imbalance. *Front. Clim.* **5**. https://doi.org/10.3389/fclim.2023.1202161. (2023).
34. Fan, C., Paynter, D., Kramer, R. J. & Lin, P. Sensitivity of Earth's radiation budget to lower boundary condition data sets in historical climate simulations. *Geophys. Res. Lett.* **52**, 2025–115914 (2025).
35. Zelinka, M. D., Smith, C. J., Qin, Y. & Taylor, K. E. Comparison of methods to estimate aerosol effective radiative forcings in climate models. *Atmos. Chem. Phys.* **23**, 8879–8898 (2023).
36. Goessling, H. F., Rackow, T. & Jung, T. Recent global temperature surge intensified by record-low planetary albedo. *Science* **387**, 68–73 (2025).
37. Allan, R. P. & Merchant, C. J. Reconciling Earth's growing energy imbalance with ocean warming. *Environ. Res. Lett.* **20**, 044002 (2025).
38. Kang, S. M. & Frierson, D. The tropical response to extratropical thermal forcing in an idealized GCM: the importance of radiative feedbacks and convective parameterization. *J. Atmos. Sci.* **66**, 2812–2827 (2009).
39. Jones, A., Haywood, J. & Boucher, O. Climate impacts of geoengineering marine stratocumulus clouds. *J. Geophys. Res.Atmos.* **114**, 10106 (2009).
40. Schneider, T., Bischoff, T. & Haug, G. H. Migrations and dynamics of the intertropical convergence zone. *Nature* **513**, 45–53 (2014).
41. Datseris, G. & Stevens, B. Earth's albedo and its symmetry. *AGU Adv.* **2**, 2021–000440 (2021).
42. Diamond, M. S., Gristey, J. J., Kay, J. E. & Feingold, G. Anthropogenic aerosol and cryosphere changes drive Earth's strong but transient clear-sky hemispheric albedo asymmetry. *Commun. Earth Environ.* **3**, 1–10 (2022).
43. McDuffie, E. E. et al. A global anthropogenic emission inventory of atmospheric pollutants from sector- and fuel-specific sources (1970-2017): an application of the Community Emissions Data System (CEDS). *Earth Syst. Sci. Data* **12**, 3413–3442 (2020).
44. IPCC: Summary for policymakers. In *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate change* (eds Masson-Delmotte, V. et al.) 3–32 (Cambridge University Press, 2021).
45. Andrews, T., Gregory, J. M. & Webb, M. J. The dependence of radiative forcing and feedback on evolving patterns of surface temperature change in climate models. *J. Clim.* **28**, 1630–1648 (2015).
46. Zhou, C., Zelinka, M. D. & Klein, S. A. Impact of decadal cloud variations on the Earth's energy budget. *Nat. Geosci.* **9**, 871 (2016).
47. Andrews, T. et al. On the effect of historical SST patterns on radiative feedback. *J. Geophys. Res.: Atmos.* **127**, 2022–036675 (2022).
48. Sciences Engineering Medicien, N.A. *Medicine: reflecting sunlight: recommendations for solar geoengineering research and research governance*. https://doi.org/10.17226/25762. https://nap.nationalacademies.org/catalog/25762/reflecting-sunlight-recommendations-for-solar-geoengineering-research-and-research-governance. (The National Academies Press, 2021).
49. Grosvenor, D. P. et al. Remote sensing of droplet number concentration in warm clouds: a review of the current state of knowledge and perspectives. *Rev. Geophys.* **56**, 409–453 (2018).
50. Von Salzen, K. et al. Reduced aerosol pollution diminished cloud reflectivity over the North Atlantic and Northeast Pacific. *Nat. Commun.* **16**, 9433 (2025).
51. Watanabe, M. et al. Possible shift in controls of the tropical Pacific surface warming pattern. *Nature* **630**, 315–324 (2024).
52. Hwang, Y.-T., Xie, S.-P., Chen, P.-J., Tseng, H.-Y. & Deser, C. Contribution of anthropogenic aerosols to persistent La Niña-like conditions in the early 21st century. *Proc. Natl. Acad. Sci.USA* **121**, 2315124121 (2024).
53. Tseng, H.-Y. et al. Fast and slow responses of the tropical Pacific to radiative forcing in Northern high latitudes. *J. Clim.* **36**, 5337–5349 (2023).
54. Stephens, G. L. et al. The curious nature of the hemispheric symmetry of the Earth's water and energy balances. *Curr. Clim. Change Rep.* **2**, 135–147 (2016).
55. Rugenstein, M. & Hakuba, M. Connecting hemispheric asymmetries of planetary albedo and surface temperature. *Geophys. Res. Lett.* **50**, 2022–101802 (2023).
56. Loeb, N. G. et al. Continuity in top-of-atmosphere earth radiation budget observations. *J. Clim.* **37**, 6093–6108 (2024).

57. Wilks, D.S. "The Stippling Shows Statistically Significant Grid Points" How Research Results are Routinely Overstated and Overinterpreted, and What to Do about It. *Bull. Am. Meteorol. Soc.* Accessed 29 July 2025. https://doi.org/10.1175/BAMS-D-15-00267.1. (2016).
58. Huang, B. et al. Uncertainty estimates for sea surface temperature and land surface air temperature in NOAAGlobalTemp Version 5. *J. Clim.* **33**, 1351–1379 (2020).
59. Lenssen, N. J. L. et al. Improvements in the GISTEMP uncertainty model. *J. Geophys. Res. Atmos.* **124**, 6307–6326 (2019).
60. Rohde, R. A. & Hausfather, Z. The Berkeley earth land/ocean temperature record. *Earth Syst. Sci. Data* **12**, 3469–3479 (2020).
61. Yuan, T., Song, H., Boss, L. F. & Diamond, M. S. Detectable ship tracks account for just 5% of aerosol indirect forcing from ship emissions. *Commun. Earth Environ.* **6**, 899 (2025).
62. Seinfeld, J. H. et al. Improving our fundamental understanding of the role of aerosol-cloud interactions in the climate system. *Proc. Natl. Acad. Sci.USA* **113**, 5781–5790 (2016).
63. Soden, B. J., Broccoli, A. J. & Hemler, R. S. On the use of cloud forcing to estimate cloud feedback. *J. Clim.* **17**, 3661–3665 (2004).
64. Van Loon, S., Rugenstein, M. & Barnes, E. A. Observation-based estimate of Earth's effective radiative forcing. *Proc. Natl. Acad. Sci.USA* **122**, 2425445122 (2025).
65. Yuan, T., Remer, L. A. & Yu, H. Microphysical, macrophysical and radiative signatures of volcanic aerosols in trade wind cumulus observed by the A-Train. *Atmos. Chem. Phys.* **11**, 7119–7132 (2011).
66. Chen, Y. et al. Substantial cooling effect from aerosol-induced increase in tropical marine cloud cover. *Nat. Geosci.* **17**, 404–410 (2024).
67. Yuan, T. et al. Analyses of virtual ship-tracks systematically underestimate aerosol-cloud interactions signals. *Geophys. Res. Lett.* **52**, 2024–114356 (2025).
68. Soden, B. J. et al. Quantifying climate feedbacks using radiative Kernels. *J. Clim.* **21**, 3504–3520 (2008).
69. Kramer, R. J., Matus, A. V., Soden, B. J. & L'Ecuyer, T. S. Observation-based radiative Kernels From CloudSat/CALIPSO. *J. Geophys. Res. Atmos.* **124**, 5431–5444 (2019).
70. Smith, N. & Barnet, C. D. CLIMCAPS—A NASA long-term product for Infrared + Microwave atmospheric soundings. *Earth Space Sci.* **10**, 2022–002701 (2023).
71. Block, K. & Mauritsen, T. Forcing and feedback in the MPI-ESM-LR coupled model under abruptly quadrupled CO_2. *J. Adv. Modeling Earth Syst.* **5**, 676–691 (2013).
72. Arola, A. et al. Aerosol effects on clouds are concealed by natural cloud heterogeneity and satellite retrieval errors. *Nat. Commun.* **13**, 7357 (2022).
73. Kramer, R. Kernel Decomposition of CERES Radiative Flux Changes. Zenodo https://doi.org/10.5281/ZENODO.17825264 (2025).

Acknowledgements

We thank Anonymous reviewers for their helpful comments. We acknowledge funding support from NASA MEaSUREs and TASNPP programs (grant numbers 80NSSC24K0458, 80NSSC24M0014 and 80NSSC24M0045), NOAA ERB program(grant NA23OAR4310299), and DOE ASR program(grant DE-SC0024078).

Author contributions

T.Y. conceived the idea, analyzed the data, and wrote the manuscript draft. H.S. processed the data and made plots. R.K., L.O., S.P.R., R.W., and M.C. contributed to the writing of the manuscript.

Competing interests

There are no competing interests to declare.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-72926-3.

Correspondence and requests for materials should be addressed to Tianle Yuan.

Peer review information *Nature Communications* thanks Richard Allan and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

Reprints and permissions information is available at http://www.nature.com/reprints

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026